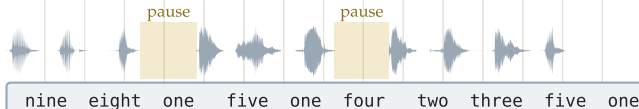


context, pinned in the system slot for the whole session

{sys} User profile — name: Priya Raman; email: priya.raman@gmail.com; plan: business tier {/sys}

(a) The usual pipeline — a separate recognizer + a silence timer

audio
(caller reads
a 10-digit no.)
recognizer
output:



transcript
tokens only

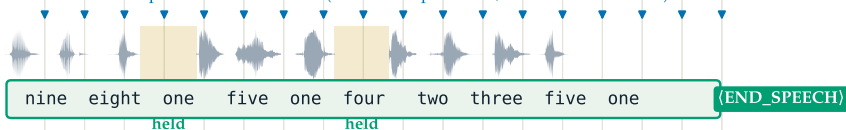
turn-end call
(silence timer):

✗ ✗ ✗ ★ patient ($X = 1.0$ s):
+0.9 s, slow
responsive timeout ($X = 0.5$ s): fires inside the pauses — 2 premature / number
the clock is blind to the digits: no single timeout is both prompt and pause-proof

(b) Ours — the recognizer emits the transcript AND the end-of-turn token, inline

one decode per 0.5 s audio chunk (≈ 0.1 s compute each, well inside real time)

audio
decode
output:



at a pause ($t = 4.5$ s)

{sys} ...priya.raman@gmail.com...

{audio 0.0–4.8 s}

{asst} nine eight one · five one four

model emits: → four (no {END})

at the turn end ($t = 7.5$ s)

{sys} ...priya.raman@gmail.com...

{audio 0.0–7.5 s}

{asst} ...two three five one

model emits: → {END_SPEECH}

can output:

any word, plus

two markers

{(EAGER_END),

{END_SPEECH}}

HOLD — the 3-3-4 pattern predicts more digits

FIRE — complete + 0.3 s silence heard, +0.42 s